

Discovering Links in Datasets with Temporal Information

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Abstract. More links in linked data make datasets semantically richer. A dataset owner usually creates links in data and makes them explicit, however, it is also possible to find links that are not explicitly present in data based on structural patterns of nodes. In this paper we present a tensor-based model for representation of RDF datasets with temporal information. We develop a novel configurable link discovery method that uses time as a relevant information for link discovery. The method can enrich existing RDF datasets with new links that are relevant at certain time. We evaluate the method on a real-world dataset from ProgrammableWeb and we present its performance and capabilities.

Keywords: linked data, link discovery, temporal information, tensor factorization

1 Introduction

Over the last few years the Linked Data Cloud has grown significantly. The number of published datasets has increased from only 12 in May 2007 and 29 in July 2009, to 295 datasets in September 2011 with over 31.6 billion triples¹. One of the key Linked Data principles is to use URI references to point to entities or individuals within or outside datasets. Such links are usually created at the time of publishing a dataset and often do not change unless a dataset owner alters them. A link discovery algorithm, on the other hand, allows to find new links in datasets that are not explicitly present but they implicitly exist due to structural patterns in datasets and the temporal information such as time when the links were created.

In this paper we present a novel configurable method for time-aware link discovery in RDF datasets. The method exploits the *temporal information* and the *structural characteristics* of the dataset. The method recommends new, previously not existing links that can be considered as relevant at certain time. We propose a tensor-based model for modelling RDF data with temporal information. We use the tensor factorization method as an underlying concept for

¹ <http://lod-cloud.net/state/>

evaluation of the existence of new links between instances in a RDF dataset. We evaluate the method on a real-world dataset and we present its performance and capabilities.

The paper is structured as follows. Section 2 describes the extended version of the Linked Web APIs used for evaluation purposes. Section 3 describes the link discovery method, its notations, definitions and the supporting algorithm. Section 4 describes several experiments to present the performance and capabilities of the method. Section 5 discusses different aspects of the method. In Section 6 we give an overview of the related work, and finally, Section 7 concludes the paper and presents the future work.

2 Extended Linked Web APIs Dataset

For evaluation purposes we created extended version of the *Linked Web APIs* dataset. The dataset is an RDF representation of the ProgrammableWeb² directory, the largest mashup and Web APIs directory. The dataset contains information about developers, mashups they created and Web APIs they used, together with categories they belong to. In addition, the dataset has information about tags assigned to each mashup and a Web API, formats and protocols that Web APIs support. We also collected information about the time when users, mashups or Web APIs appeared in the directory for the first time. Figure 1 shows an extract of the extended Linked Web APIs dataset.

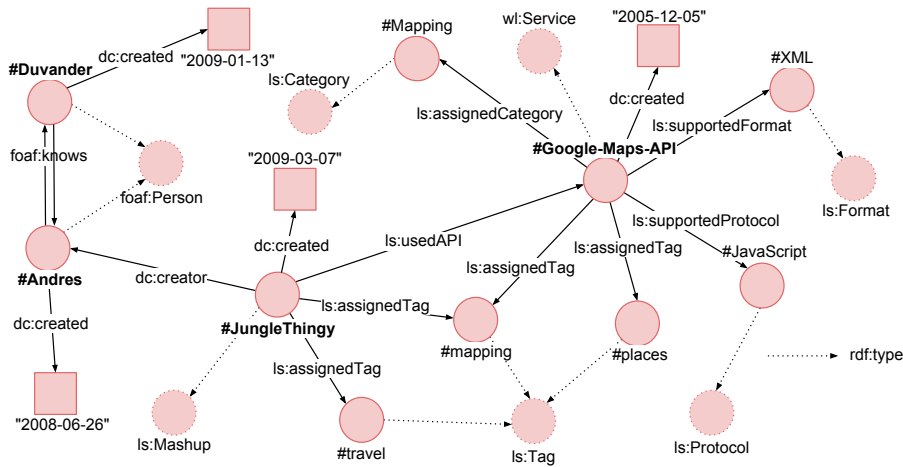


Fig. 1: Excerpt from the extended Linked Web APIs dataset

The dataset uses several well know ontologies and vocabularies: FOAF³ ontology (*prefix foaf*) - concept *foaf:Person* describes users and property *foaf:knows*

² <http://www.programmableweb.com/>

³ <http://xmlns.com/foaf/spec/>

describes a social relationship between users, WSMO-lite [17] ontology (*prefix wl*)- concept *wl:Service* describes Web APIs, Dublin Core⁴ vocabulary - property *dc:creator* describes relation between a user and a mashup, and property *dc:date* indicates creation date of a mashup or a Web API, SAWSDL [9] vocabulary (*prefix sawsdl*) - property *sawsdl:modelReference* describes a tag or a category of a Web API or a mashup. Additionally, we create new concepts and properties (*prefix ls*): *ls:Protocol* that identifies a protocol, *ls:Format* that identifies data format, and *ls:Tag* and *ls:Category* which identify a tag or a category respectively. We also create following new properties: *ls:usedAPI* - between concepts *ls:Mashup* and *wl:Service*, *ls:supportedFormat*, *ls:supportedProtocol* - between concepts *wl:Service* and *ls:Tag* or *ls:Category*, *ls:assignedTag* and *ls:assignedCategory* - between concepts *wl:Service/ls:Mashup* and *ls:Tag/ls:Category*.

3 Time-Aware Link Discovery Method

In this section we formally describe the link discovery method, its notations, definitions, and the algorithm.

3.1 Definitions

Tensor. A multi-dimensional array of numerical values [8]. The order of the tensor is the number of dimensions (ways, modes) that the tensor uses. In our method we use a tensor of order three denoted by $\mathcal{Y}^{I \times J \times K}$. The (i, j, k) element of a third-order is denoted as y_{ijk} .

Information Ageing. A process of retention of information in a memory over time. We represent the relation between time and retention using a forgetting curve[3]: $R = e^{-\frac{T}{S}}$ where R is the memory retention, T is the amount of time since the information was received and S is the strength of the memory.

Based on the definition of the forgetting curve, we propose an ageing function

$$\mathcal{A}(t) = \mathcal{A}(t_o) * e^{-\lambda t} \quad (1)$$

where $\mathcal{A}(t)$ is the amount of information at time t , $\mathcal{A}(t_o)$ is the amount of information at present, λ is ageing/retention factor and t is the age of the information (i.e., the retention period).

We use two configurable parameters that influence the ageing process. The λ parameter influences the strength of the memory. The higher the value of the λ parameter is, the faster the information retention is. Similarly, the older the information is, the lower is the amount of held information.

⁴ <http://dublincore.org/documents/dcmi-terms/>

3.2 Modelling RDF Datasets with Temporal Information

We use tensors to model a relational data with temporal information described in RDF. In order to model the RDF dataset with more then one type of relations as the tensor, we use third-order tensor notation proposed in [11]. Let $\mathcal{Y} \in \{0, 1\}^{N \times N \times M}$ be a tensor representing an RDF dataset. The tensor consists of two identical dimensions N representing nodes (instances) in the dataset, and third dimension M representing the relations (predicates) that explicitly exist in the dataset. The N is the number of instances and M is the number of relations (predicates) in the RDF dataset. The tensor element $y_{ijk} = 1$, if the i -th instance is in k -th relation with the j -th instance, for $i, j \in \langle 0, N \rangle$ and $k \in \langle 0, M \rangle$. Otherwise, the tensor element $y_{ijk} = 0$.

Each tensor element in the model has value of 1 or 0 if a relation between the two instances exists or not. In this paper we propose extension of this representation with using of the temporal information. In other words, the values of the tensor elements are gradually reduced with respect to the time when the corresponding relation (predicate) appeared in the dataset. Let $\mathcal{X} \in \mathbb{R}^{N \times N \times M}$ be a modelled tensor at time t . We then compute the value of x_{ijk} tensor element by using the ageing function as

$$x_{ijk} = y_{ijk} * e^{-\lambda t} \quad (2)$$

where y_{ijk} is the initial value of the tensor element ($\{0, 1\}$), λ is the ageing factor and t is the link's age. See Section 3.1 for additional details about the ageing function.

For example, consider a RDF dataset consisting of 4 instances of concept *foaf:Person* (c_1, c_2, c_3, c_4), and 3 predicates *foaf:knows* indicating friendship relation, ($c_1 \xrightarrow{t_1} c_2, c_3 \xrightarrow{t_1} c_2, c_2 \xrightarrow{t_{15}} c_4$). The t_x indicates the age of the link in weeks. For example, in $c_2 \xrightarrow{t_{15}} c_4$, it indicates that the relation was established 15 weeks ago since the present t_0 .

Table 1 shows this information modelled as a tensor both with and without ageing. Note that the link between the person c_2 and c_4 has a lower value due to the fact that this link appeared earlier then the other two.

	c_1	c_2	c_3	c_4
c_1	0	1	0	0
c_2	0	0	0	1
c_3	0	1	0	0
c_4	0	0	0	0

(a) Tensor model without ageing

	c_1	c_2	c_3	c_4
c_1	0	0.99	0	0
c_2	0	0	0	0.86
c_3	0	0.99	0	0
c_4	0	0	0	0

(b) Tensor model with ageing

Table 1: Example of modelling data

In our dataset we do not have explicit information about time when links were created thus we assume that a link of a triple was created at the same time

when the subject or the object of the triple. In other words, the creation time of a link is the same as the creation time of a Web API or a mashup that the link is associated with.

3.3 Uncovering Hidden Latent Factors

We use the tensor factorization as an underlying concept to perform structural analysis of a RDF dataset over time. We propose extension of the RESCAL approach [11] with employing the time as an important component. Each frontal slice $\mathbf{X}_k$ of a tensor is factorized as

$$\mathbf{X}_k \approx \mathbf{A}\mathbf{R}_k\mathbf{A}^T, k = 1 \dots m \quad (3)$$

where $\mathbf{A}$ is $n \times r$ matrix which models the participation of an instance in latent factor r , and $\mathbf{R}_k$ is $r \times r$ matrix that models the interactions of the latent factors for the k -th predicate. The r is configurable parameter of the factorization algorithm, which indicates the number of latent factors to be uncovered.

The matrix $\mathbf{A}$ and the matrices $\mathbf{R}_k$ are computed by solving the minimization problem

$$\min_{\widehat{\mathbf{X}}_k} \|\mathbf{X}_k - \widehat{\mathbf{X}}_k\|_F, \text{ where } \widehat{\mathbf{X}}_k = \mathbf{A}\mathbf{R}_k\mathbf{A}^T \quad (4)$$

For computing the factorization we employ the RESCAL algorithm [11] and adapt in our model. Although there exists another tensor factorization algorithms, RESCAL is suitable for analysis of multi-relational data and link prediction task. In particular, our variant of the algorithm uses as input a tensor, where the elements are real positive numbers, instead of only $\{0, 1\}$ as in the RESCAL algorithm (see Section 3.2 for discussion on how we model a tensor).

3.4 Time-Aware Link Discovery

The *link discovery* task is described as a task to evaluate the possible existence of links between two pair of instances. We focus on *temporal link discovery* problem, where the goal is to discovery new, previously not present links, while taking into the account temporal information about the existing links in the dataset. In other words, we consider analyzing the structure of the dataset across the time.

Our developed method uses tensor factorization to analyze the structure of a RDF dataset. The method uses the results from the analysis to recommend new links. To evaluate existence of a link between i -th and j -th the instance we do reconstruction $\widehat{\mathbf{X}}_k = \mathbf{A}\mathbf{R}_k\mathbf{A}^T$ of the appropriate slice $\mathbf{X}_k$ for the k -th predicate.

The algorithm for time-aware link discovery is as follows.

1. Model tensor $\mathcal{X}$ for the input RDF dataset and ageing constant λ .
2. Compute tensor factorization for the tensor $\mathcal{X}$ with the adapted RESCAL algorithm.

3. Reconstruct frontal slice $\widehat{\mathbf{X}}_k$ using the latent factor $\mathbf{R}_k$ and matrix $\mathbf{A}$, where k indicates the predicate in question.
4. Read entry values x_{ijk} for i -th row (instance in question) and each j -th column. The entry values indicate whether a link between the i -th instance in the question and instance in the j -th column should exist. The higher the values is, the higher the confidence for existence of the link is.
5. Sort the entry values in decreasing order and return highest ranked ones. The top- N ranked are instances that should be linked with the instance and link type in question.

Note that the link prediction (see step 5 from the algorithm) can be also done by comparing $x_{ijk} > \theta$ to some threshold θ .

4 Experiments

In this section we demonstrate the proposed link discovery method on a real-world dataset from ProgrammableWeb introduced in Section 2. The dataset contains information from June 2005 till the end of March 2013 and has in total 22286 of instances and 8 types of relations. The constructed full tensor model has following dimensions $22286 \times 22286 \times 8 \approx 4 \times 10^9$. In the model can exist approx. 4×10^9 relations between all instances. The real-world dataset contains approx. 123×10^3 relations at the end of March 2013.

For the purpose of analysing the data over different time periods we prepared snapshots of the dataset in various times. All snapshots are prepared in cumulative manner. The i -th snapshot contains data from the previous snapshot $i - 1$ and data from the new step. First snapshot contains data from June 2005 till the 1st of January 2008. We selected this time period for the first snapshot in connection to the ageing constant setting and in connection to the number of relations in this time period. In the first snapshot we have approx. 21×10^3 relations. It is significant portion of total number of relations and should contain sufficient information for the link discovery. The step for the next snapshot is three months, and it contains all data till April 1st 2008. The last snapshot contains all data until April 1st 2013. In total we prepared 22 snapshots modelled as tensors, with and without ageing. As present for each snapshot we take the end time of the snapshot.

We constructed both sets of snapshots to compare the capabilities of our proposed time-aware link discovery and the link discovery without time information.

The setting of ageing function for our experiments is: ageing constant $\lambda = 0.01$ and the age period t is calculated in weeks. The dataset covers information for more than seven years, it is approx. 360 weeks. The first snapshot covers approx three years (150 weeks). Figure 2 depicts the level of influence of the ageing function for different λ .

For the experiments we implemented own version of RESCAL [11] tensor factorization algorithm in R⁵. The tensor factorization has configurable parameters

⁵ <http://www.r-project.org/>

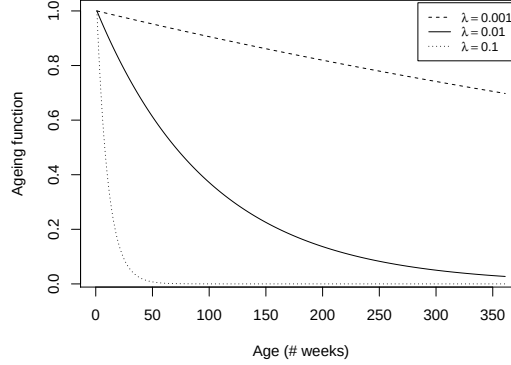


Fig. 2: Ageing function

in order to influence the minimization of error and to influence the computation time. For our experiments we experimentally set the number of latent factors to 40 and stopping condition for the minimization task is: if change of the factor matrices between two iterations is < 1 .

The questions we address in the experiments are:

- *How the ageing function influences the link discovery?*
- *How to discover new links in different periods of time?*
- *How the structure of the dataset in different time intervals influences the link discovery?*

4.1 Relation of Discovered Links and Time Information

In this experiment we present the relation of discovered links and time information of entities, which participate in the discovered links. We focused on the reconstructed values as the result of tensor factorization. We assume that the reconstructed values reflect the time - the higher the reconstructed value for the new link is, the more the new link is influenced by the existing relations in the recent time.

Our dataset contains temporal information for each Web API - the time the Web API was added to the directory. The goal of this experiment is discovering new links, where the subjects of discovered links are Web APIs and the object is one specific tag. The particular example is: Which Web APIs should also have the tag *mapping*?

Table 2 presents the top 10 Web APIs (their names and dates, they were added to directory), which should also have the tag *mapping*. For this experiment we used the last snapshots with and without ageing, which contains all data till April 1st 2013. The setting of presence is set as April 1st 2013 too. In column *No ageing* are listed the top 10 Web APIs on the dataset, where the ageing function was not used. We computed this results to compare the link discovery with and without considering the time. In column *Ageing* is the set of top 10

Position	Without Ageing		With Ageing	
	Name	Date	Name	Date
1	Placr	2011-07-07	SetGetGo IP Geolocation	2013-12-29
2	BestParking	2011-01-06	JetSetMe	2012-11-11
3	Tube Updates	2011-03-27	Frontier Airlines Word Wheel Local	2013-02-10
4	ParkWhiz	2010-09-30	Eaupen	2012-12-29
5	Eaupen	2012-12-29	DC Location Verifier	2012-10-17
6	NAVTEQ Traffic	2011-09-11	TripCheck	2013-03-02
7	Jeppesen Journey Planner	2011-10-26	View	2013-03-01
8	View	2013-03-01	WikiSherpa	2012-07-06
9	MyTTC	2011-08-13	ThinkGeo Cygnus Track	2012-10-26
10	Pearson Eyewitness Guides	2011-09-16	weather-api.net	2012-12-19

Table 2: Top 10 APIs which should have tag *mapping*, with/without ageing on the 1st of April 2013

Web APIs computed on the snapshot with ageing - these Web APIs should be affected by the existing links, which appeared much more close to the present time in the dataset. In other words, the time information of these Web APIs should be much more close to the present (April 1st 2013). The results supports and verify our assumption, that the higher reconstructed values leads to the links between entities with time information much more close to the present than the link discovery without time information.

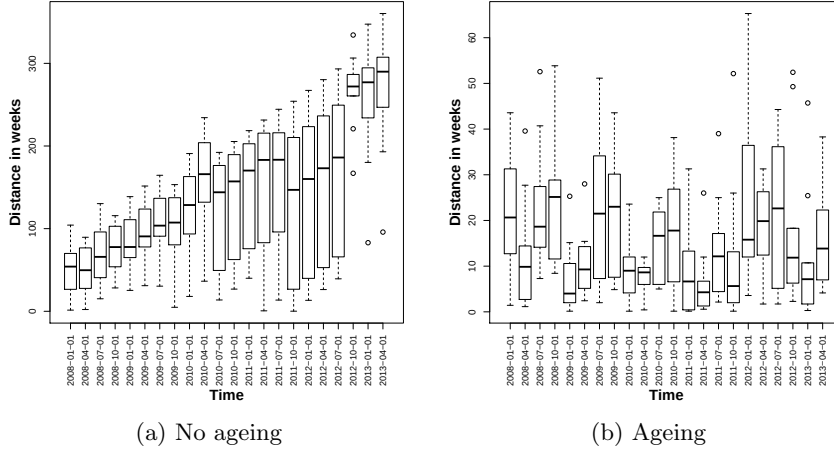


Fig. 3: Distance of discovered APIs from the present

The same experiment we run also for all 22 snapshots. Figure 3a and Figure 3b depict the distance in weeks of top 10 Web APIs from the present in different snapshots. We use the standard box plot to examine the distributions of distances graphically. The distance is measured in weeks for top 10 Web APIs in each snapshot. Figure 3b presents much more lower distances than the Figure 3a. This supports our assumption, that the higher the reconstructed value is, the more it was influenced by the data in recent time.

4.2 Significance of Discovered Links Over Time

In this experiment we present how can the time information influence the significance of the discovered link. The significance is expressed as reconstructed value from results of tensor factorization. The higher significance reflects the position in the discovered set of links.

This experiment finds the set of links where the objects are tags and subject is API. The more specific description: Which top N tags should be assigned to the Google Maps API?

Position	Without Ageing	With Ageing
1	travel	geolocation
2	realestate	location
3	sports	travel
4	reference	government
5	uk	geocoding
6	location	visualization
7	transit	transportation
8	food	gis
9	science	weather
10	government	deadpool

Table 3: Top 10 significant tags for Google Maps API on the 1st of April 2013

Table 3 show the results of experiment with and without ageing - top 10 tags for Google Maps API. The *Position* represents better significance (higher reconstructed values). For this example we used the snapshot from April 1st 2013 with and without ageing. Some of discovered tags are the same in both sets, but on different positions. The column *No ageing* in Table 3 contains the discovered links to tags - this set is influenced by the hidden factors in the whole dataset, since the snapshot with all relations without ageing is used. The column *Ageing* contains the set of tags influenced by time, since the snapshot with ageing is used. Based on our assumption and result of previous experiment, the second set represents the set of tags that are more influenced by the recent relations in the snapshot.

Figure 4 depicts the positions (better positions are more significant) of tags in top 10 set for *Google Maps API* over time. The position is represented by the color. The darker the box is, the better is the position of tag. Figure 4a depicts the positions for the case the ageing is not applied. In the visualization you can see the tendency "stability" of positions over time. It is caused by the existence of links in global. The shade of gray is "stable" from left to the right. Figure 4a depicts the positions for the case the ageing is applied. The subset of tags were on better position in the past - darker boxes in bottom left corner. The new subset of tags is more preferred in recent time - darker boxes in top right corner.

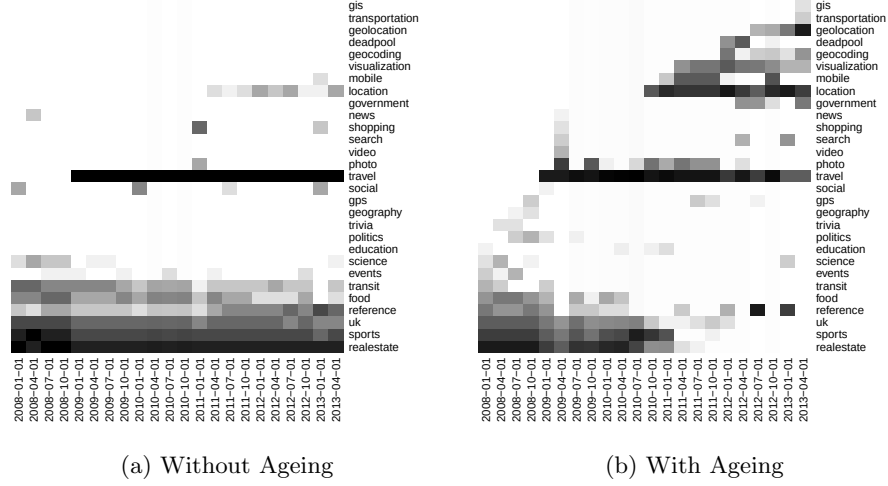


Fig. 4: Visualization of positions for each snapshot

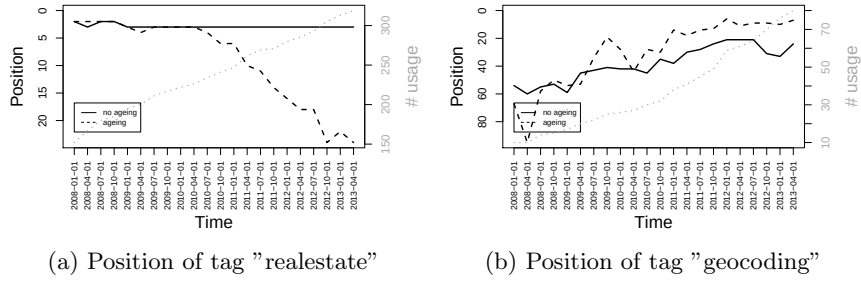


Fig. 5: Evolution of position over time for specific tag

We also focused on specific tags and their discovered position in different snapshots over time. Figure 5a depicts the evolution of the position for two tags with respect to time. On the same figure we present the usage of the tags.

In case there is no ageing used, the position of the tag *realestate* is stable, while with ageing the position is getting lower. It means, it is not preferred in recent time. The Figure 5b depicts the same information for the tag *geocoding*. Here, the evolution of position is similar for both cases with and without ageing. The position is better together with the number of usages grows.

The results from this experiment show that the importance of discovered links can change over the time. In this experiment we demonstrate that for example the tag *realestate* is gradually losing its importance. While the tag *geocoding* is slowly gaining the importance. In the next experiment we look into the causes for such behaviour.

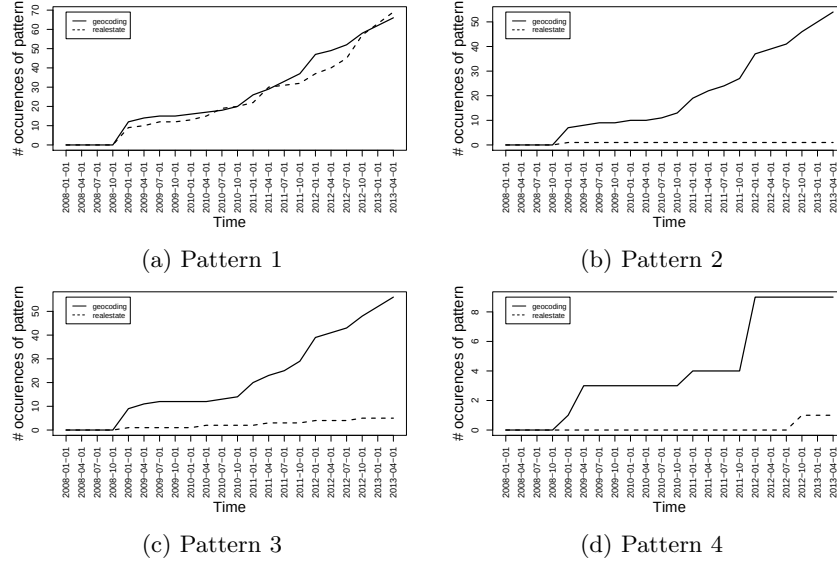


Fig. 6: Number of occurrences for each pattern

4.3 Impact of Dataset Structure

In this experiment we demonstrate how the tensor factorization with ageing takes into account structure evolution when discovering new links.

As we proposed, the approach we use is based on identification of hidden patterns in structure of data (tensor factorization) in connection to the time information (ageing). Identified hidden patterns are used to discover new links in data. To identify such significant patterns we can use the existing local property of graphs, called Motifs. Motifs are defined as recurrent and statistically significant sub-graphs.

We adopted the idea of motifs in this experiment as the "evidence" of influence of structure and time information in tensor factorization with ageing. This experiment demonstrates and provides explanation for the results from previous experiment: why is not the tag *realestate* on the upper position in recent time and why is growing the position of tag *geocoding*.

The new links for the Google Maps API can be discovered only in case the similar pattern exists in data and the pattern contains information related or is similar to the Google Maps API structure. We define several patterns, which might influence the link discovery for the tags *realestate* and *geocoding*.

We define following patterns:

1. *?var rdf:type wl:Service AND ?var ls:assignedTag ls:realestate*
2. *?var ls:assignedCategory ls:Mapping AND ?var ls:assignedTag ls:realestate*
3. *?var ls:assignedTag ls:mapping AND ?var ls:assignedTag ls:realestate*
4. *?var ls:supportedProtocol ls:JavaScript AND ?var ls:assignedTag ls:realestate*

More complex pattern can exist as combination of proposed simple patterns, however, for the matter of simplicity we decided to focus only on the simple ones.

In the experiment we measure the frequency/number of occurrence of previously defined patterns for both tags and we compare them.

Figures 6a-6d depict the number of occurrences for each pattern over time. The number is "in average" significantly higher for the case with tag *geocoding* than for tag *realestate*. This leads to higher support for tag *geocoding* in link discovery with ageing. The support of tag *realestate* is getting lower because of existence of other patterns with more occurrences in link discovery with ageing.

In the case without ageing, the tag *realestate* has higher support. This is caused by the existence of patterns in the dataset from the historic perspective. These patterns do not reflect the time information. As we can see, our proposed approach promotes the patterns, which have significant number of occurrences in recent time.

5 Discussion

Robustness. The RDF dataset we used in this paper to discovery links is created from the data collected from the service directory ProgrammableWeb. Although the method was evaluated on a domain-specific dataset, our developed method is enough robust to discover new links in a dataset from any other domain. The only prerequisite is existence of temporal information when the links were created. It is usual that the links are created at the same time when the object was created. For example, a link between mashup and a user is created at the same time when the user creates the mashup. Another possibility is to create incremental snapshots of a given dataset and to monitor the changes.

Structural Patterns. The results from the link discovery are highly dependent on the structure of the RDF graph and the time when the links appeared. In Section 4.3 we identified simple structural patterns that influence the link discovery in this specific dataset. However, there is no reason to assume that there cannot be present also other, more complex structural patterns that influences the link discovery. In our future work we plan to explore methods for automatic detection of more complex patterns.

Snapshot Creation. The length of the snapshots was chosen so that the snapshots have sufficient amount of data for learning. Note that the data of some snapshots can be differently distributed with respect to the time. Some snapshots might have the data normally distributed, while in some the majority of the data might be at the beginning or at end of the snapshot. Such distribution of the data, of course, has different impact on the link prediction. For example, when the majority of the data is at the beginning, the link discovery will be influenced by the historical data.

Link Significance. Currently, we used the method for discovering new, previously not present links. However, the method can be used to evaluate validity of existing links. Instead of, looking at top-N links with highest confidence value, we can look at the top-N links with the lowest value. Between those links, might be also links that are present in the dataset, which might be considered as no more valid and be removed. In our future work we want to evaluate this aspect of the method and do research on the confidence intervals for new and old, no more valid links.

6 Related Work

There are two main topics closely related to our time-aware link discovery method: tensor factorization and relational learning. The models and methods covered by these topics are used to model multi-relational data and to perform the link discovery.

Most researches in relational learning are based on statistical relational learning. These approaches are build upon the Bayesian or Markov networks [5,7] or combinations with tensor representations [6].

There is a growing interest in tensor models and factorizations in multi-relational data modelling. The growing interest is affected by the natural representation of multi-relational data using dimensions of tensors. The overview of tensor factorizations and applications is in [8]. We adopted the model by [11], where each frontal slice of tensor represents each relation. Similar model was also used in [10]. Both modelling approaches do not incorporate the time information. They only take into account entities and relations among them. These methods are called link-information-based approaches. The other side is node-information-based approaches, they take into account attributes of entities [16,14]. The modification in [12] is able to work with attributes (time attribute can be also included) and combine both approaches. But they do not consider the impact of time.

Our link discovering method incorporates time information as one of the key aspects. Our method partially combines link-information-based and node-information-based approaches. There are existing researches, that use time for predicting links. In [15] is used third-order tensor factorization, two dimensions are used to represent relations and the third is time. This approach is suitable only for one type of relation. Similarly in [1,2,4]. It can be use only for dataset with one type of relation. Our method is focused on multi-relational data with time information.

There are also other approaches using temporal information for link prediction, e.g. prediction links in asynchronous communication in Social Networks [13], which are less relevant.

7 Conclusion and Future Work

The popularity of publishing RDF datasets as Linked Data is significantly growing in recent time. One of the most important principles when publishing data as Linked Data is providing links. The set of links in datasets might be incomplete or some links might be no more valid. Although there exists number of sophisticated approaches for link discovery, there is still a lack of a methods that will incorporate the time information which should improve the performance of the link discovery and provide more relevant links. In this paper we proposed time-aware link discovery method that extends the tensor factorization to solve link discovery task. The method exploits the temporal information found in RDF datasets to discover new links in different time periods. We proposed tensor-based model for modelling multi-relational data. While existing methods use either multi-relational data or data with one type of relation and time information, our method utilizes both, the multi-relational data and time information. The results from the experiments, run on a real-world dataset, show that the method exploits the structure of the datasets and promotes links recommended as result of those structures. The results also show that the method takes into account the temporal information and discovers links closer to the recent time.

In future we want to improve the detection of structural patterns and their combinations. These patterns should help us better identify the behaviour of the method. We also plan to investigate other ways of incorporating the temporal information. Further, we plan to explore different applications of the method. In particular, we want to evaluate the performance of the method in evaluation of existing links. Last but not least we want to evaluate the method on datasets from the Linked Data cloud.

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